SEAGAS - Team Update: What Is Done, What Is Left, What To Do Next

From Carl (Soh Way Miin), Backend Developer, to Group 10 | 28 October 2026
COS40005 Computing Technology Project A | POC004

In short: every advisor is now a **Dean** or a **Head of Programme** with a 5-digit **staff ID**, and only sees and adds students of their own courses. Advisors get their login details by email, like students. The Business course now has its own **skills, role requirements and learning resources**, students only see the skills of their course, and every student has a **programme** (major). After you pull, run **start_all.bat** once.

Note: the new-student list now needs a **Programme** column, and advisors you created before became Dean of Computing (or Head of Programme, CS), so they no longer see Business students. Please read section 2 before testing.

1. What Is Done (28 October)

1.1 Advisor positions, courses and staff IDs

Position	Sees and adds	Staff ID
Dean, Computing	Computer Science and ICT students	`10xxx`
Head of Programme, CS	Computer Science students	`11xxx`
Head of Programme, ICT	ICT students	`12xxx`
Dean, Business	Business students	`20xxx`
Head of Programme, Business	Business students	`21xxx`

Area	What changed
One Dean per office	Computing and Business each have only one active Dean; Heads of Programme are not limited. To replace a Dean, change the old Dean's position or deactivate the account first. The Add / Edit form shows who the current Dean is.
What advisors see	Student list, overview numbers, skill-gap chart, student detail and evidence downloads only cover their courses (another course's student is "not found"). The course buttons only show their courses. New-student lists: rows of other courses are skipped, and the Excel template only offers their courses.
Creating an advisor	The admin enters name, **personal email** (from HR), position and staff ID (a free one is suggested). The school email is `<staff ID>@advisor.newinti.edu.my` unless changed. The system makes a temporary password and emails staff ID, school email and password to the personal email; the advisor must change the password at first sign-in. Without a mail server the email is in **Student Intake → Email outbox**. Admin accounts are still created with a typed password.
Editing users	Admins can change name, **school email** (domain must match the role), position and staff ID, and reset passwords.

1.2 Business skills, roles and learning resources

Area	What changed
29 Business skills	Finance (Financial Modelling, Accounting Principles, Financial Reporting, Budgeting & Forecasting ...), marketing (Digital / Social Media Marketing, SEO, Market Research, CRM ...), business analysis (Business Process Modelling, KPI Reporting ...), HR (Recruitment, Employment Law, Payroll, Performance Management ...), plus Negotiation and Business Writing. They use the existing categories (domain skills count as Technical, the area is in the subcategory), so the 7 readiness dimensions stay the same.
Business roles	Business Analyst, Marketing Executive, Financial Analyst and HR Executive now have 22-23 skills each (required / preferred / bonus) and are seed roles, like the 6 IT roles. Business students can now be assessed against them.
56 learning resources	Coursera (Wharton, UVA, Minnesota ...), HubSpot, Google Skillshop, Salesforce Trailhead, Microsoft Learn, IIBA, SHRM, and Malaysian sources (MASB / MFRS, Employment Act, KWSP, PERKESO, LHDN, HRD Corp, Bursa Malaysia). Mostly free; every link was checked on 28 October.
Skills per course	Students only see the skills of their course: soft skills and tools the Business roles also use (Excel, SQL, Power BI ...) are in every list; Business-only skills for Business students; IT-only skills for CS and ICT students. ICT roles share the IT skills and resources with CS (all already covered).

All of it is in the new file `database/seed_business_skills.sql`, which start_backend.bat loads after the other seeds.

1.3 Programmes (majors)

Course	Programmes
Computer Science	Software Development, Data Science, Artificial Intelligence, Cybersecurity, Cloud Computing; Bachelor of Software Engineering
ICT	Software Systems, Data Management, Network Security, Cloud and Networking
Business	Business Analytics, Marketing, Finance, Human Resource Management

The new-student list needs a **Programme** column; in the Excel template it is a drop-down that follows the Course chosen in the same row. The admin can change it on Student Intake, it is saved on the profile when the account is created, and it is in the welcome email. Students cannot change it (shown read-only on My Profile); admins can in User Management. Students that had no programme (e.g. accounts created earlier from intake) got a random programme of their course.

1.4 Database and tests

Change	Why
`users.advisor_position`, `advisor_courses`, `staff_number`	Dean / Head of Programme, the courses they see, the 5-digit staff ID (checked against the position).
`programmes` table, `student_intake_rows.programme`	The programmes of each course; the programme of each uploaded student.
`skills_library.courses`	Which courses' students see the skill.

New backend files: `advisor_scope.py`, `programmes.py`. New endpoints: `GET /api/admin/staff-number`, `GET /api/students/programmes`. Tests: 91 pass (`python -m pytest tests -q`).

2. What To Do Next

2.1 After you pull (every member)

1. `git checkout dev`, `git pull`, then double-click **start_all.bat**. It adds the new columns and tables, loads the Business skills and resources, gives your existing advisors a position and staff ID, and gives students without a programme one.
2. Your existing advisor is now **Dean of Computing** (CS and ICT only). To see Business students, sign in as admin and either change that advisor's position in User Management, or add a Dean of Business (use a made-up personal email, then open the login email in **Student Intake** → **Email outbox**).
3. Download the new template on **New Students** (the old one has no Programme column).
4. Try a Business student (`student0002@synthetic.example.com` / `900000002` / `Seagas@2026`): run a Financial Analyst assessment and open the recommendations.
5. Use only made-up names, emails and phone numbers (NFR-05).

2.2 Before your next push

1. Pull dev first and merge it into your branch. These files changed a lot:

File	Most likely affects
`frontend/src/pages/AdminUsers.jsx`, `AdminIntake.jsx`, `AdvisorDashboard.jsx`, `AdvisorNewStudents.jsx`, `Profile.jsx`	Ren Hang, Shanice
`backend/routers/admin.py`, `advisor.py`, `intake.py`, `students.py`, `auth.py`	Anyone changing the backend
`database/schema.sql`, `migrate_existing_db.sql`, new `seed_business_skills.sql`	Yong Hin, Aaron, Carl

2. Scripts or SQL that insert advisors get a position automatically (Dean of Computing, or Head of Programme CS if that Dean exists).
3. Advisor endpoints must keep filtering by the advisor's courses (`advisor_scope.courses_of`).
4. Run the tests and `npx vite build`; push through a pull request, commit messages in English.

2.3 Known limitations and questions

- **Yong Hin:** "Financial modelling" is the name of a new skill and also an alias of the existing Financial Analysis skill, so a job description may be matched to Financial Analysis. Please decide whether to remove that alias.
- **Yong Hin:** the synthetic Business students still have IT skills from the generator, so their Business-role scores are low. The generator needs Business skill profiles.
- The programme names follow the synthetic data, not INTI's official list; send Carl the official names to replace them.
- Advisors sign in with email and password only (no staff ID on the login page).

3. What Is Still Left

Owner	Task	When
Carl	**Send real emails** (SMTP settings in `.env`, not git; "resend email" button). Advisor report export, readiness histogram, cohort progress view.	Sprint 3
Yong Hin	Business skill profiles in the synthetic data generator; the alias question above; gold labels and precision / recall / F1; 30 JDs per role (now also Business roles).	Sprint 3-4

Ren Hang, Shanice	Check the new User Management (advisor form), Student Intake (programme), New Students and My Profile pages.	Sprint 3
Aaron	Check the Business learning resources and their cost tiers; advisor notes / human oversight; Career Paths (FR-15).	Sprint 3
Not assigned	Admin System Settings (FR-16); optional CV upload; ICT-specific resources if wanted (e.g. CCNA, IT support).	Sprint 3-4
Whole team	SUS usability test; final report and demo.	Sprint 4